# VR Graffiti



Why Learn Graffiti in VR?

# Graffiti is a physical skill, not just a visual art.

Learning graffiti requires coordination between:

### Movement

controlling hand trajectories and smooth strokes.

### Distance & Speed

adjusting spray width, opacity, and consistency.

### Lettering

understanding letter structure, proportions, and styles.

# The problem

Beginners face barriers to practicing consistently, including:




**Limited access to appropriate spaces**




**Material costs**




**Safety considerations**

## The opportunity

VR can provide an accessible, repeatable environment for learning and experimenting with graffiti techniques.

Graffiti combines artistic expression with embodied motor skills, but beginners rarely have access to structured, repeatable practice with feedback.

# Existing Solutions & The Gap



## Kingspray Graffiti VR

Realistic spray physics, multiple caps, multiplayer painting, and reference projection.

**Tutorials focus on application features rather than graffiti techniques or artistic fundamentals.**

**Gap:** Primarily a creative simulation tool, better suited to experienced artists than complete beginners.



## Vermillion VR

Realistic oil painting, guided tutorials, reference tracing, and color mixing.

**Gap:** Focuses on traditional painting rather than graffiti-specific techniques and lettering.

### Our Opportunity

A beginner-oriented VR learning environment with structured instruction, progressive skill development, and measurable feedback.

# From Simulation to Skill Learning

## Observe → Trace → Reproduce → Remix

Our learning experience draws on three educational theories:

### Observational Learning

Observe and imitate an expert's hand movements through an animated virtual hand.

### Scaffolding

Gradually remove visual guides and assistance as learners gain proficiency.

### Constructionism

Develop understanding by creating personalized graffiti and experimenting with different styles.

**From imitation to independent creation.**

# Learning the Tools

Different tools, different techniques.



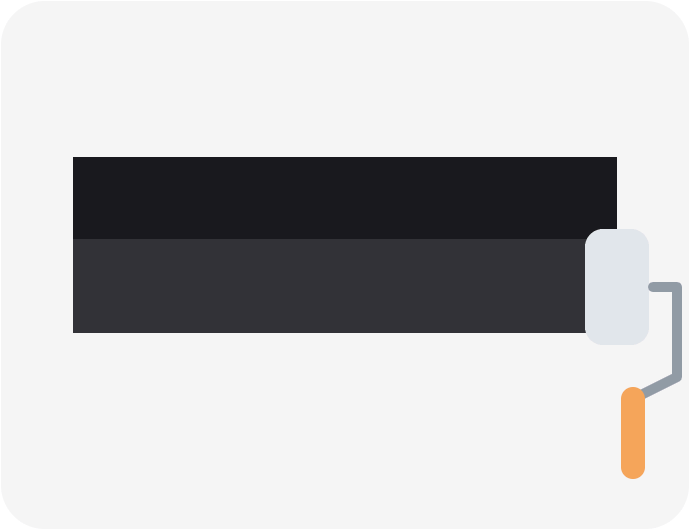
## Spray Can

Practice spray distance, movement speed, and coverage.



## Marker

Practice controlled lines, sketching, and precise outlines.

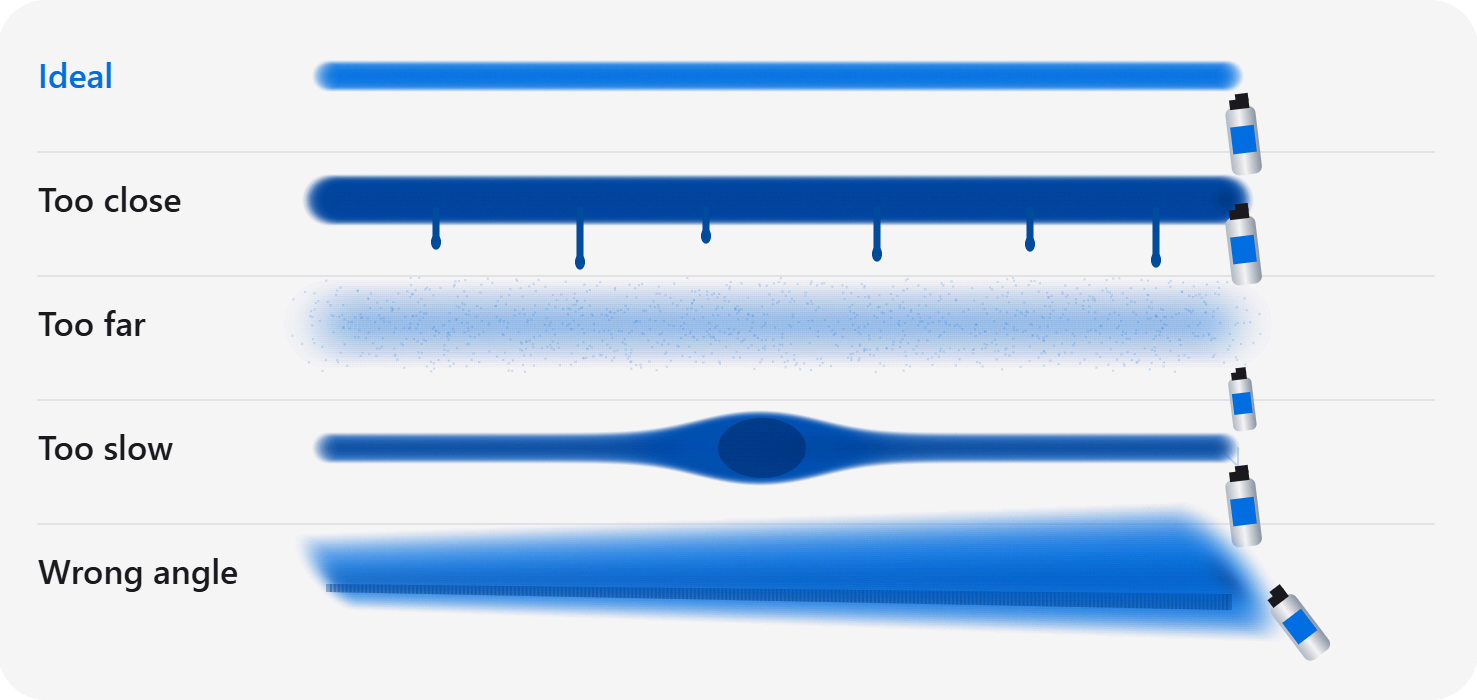


## Roller

Practice broad strokes, large fills, and surface coverage.

# Practicing Stroke Techniques

Consistent lines require precise control of distance, speed, and angle.




Same stroke. Different physical control.

## Precise control

### Distance

Coverage & edge quality

### Speed

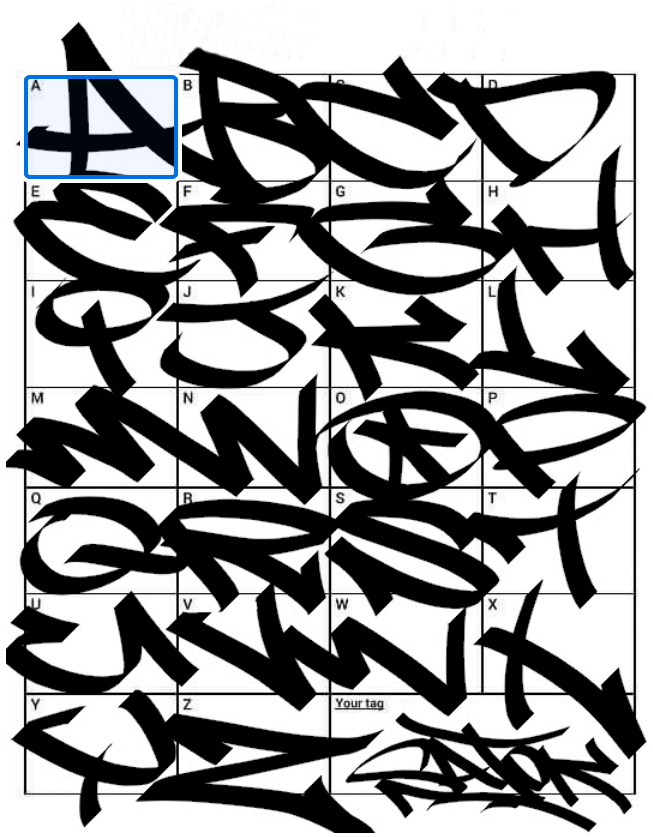
Paint buildup

### Angle

Even distribution

# Learning Lettering Styles

One letter at a time.

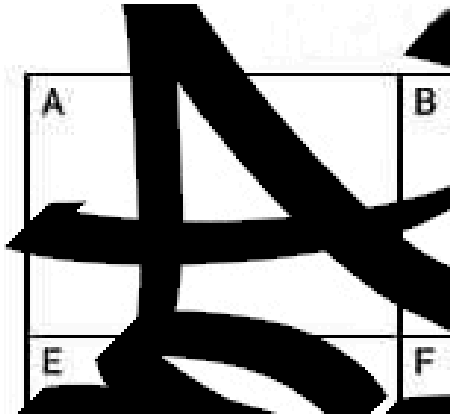



Select a letter to explore

## Letter A

Bombing Science

Explore lettering styles ↗




**Observe → Trace → Reproduce**

Structure & proportions

Stroke order & direction

Individual letters, then words and tags

# Creative Expression

From imitation to personal style.




Combine tools, strokes and letters. Create your own tag.

01 Tools → 02 Strokes → 03 Lettering → 04 Create

# Technical Implementation Plan

## Hardware & Software

**Meta Quest 3S**

Unity

C#

XR Interaction Toolkit

## Core Implementation

### Track movement

Track hand position, angle, speed, and distance from the wall.

### Simulate spray

Simulate realistic spray behavior based on movement.

### Record & replay

Record and replay expert movements for guided practice.

# A first lesson: Bubble S

From guidance to independence.




Trace with spatial cues that gradually fade

## Observe

See pose and pace.

## Trace

Follow the motion trail.

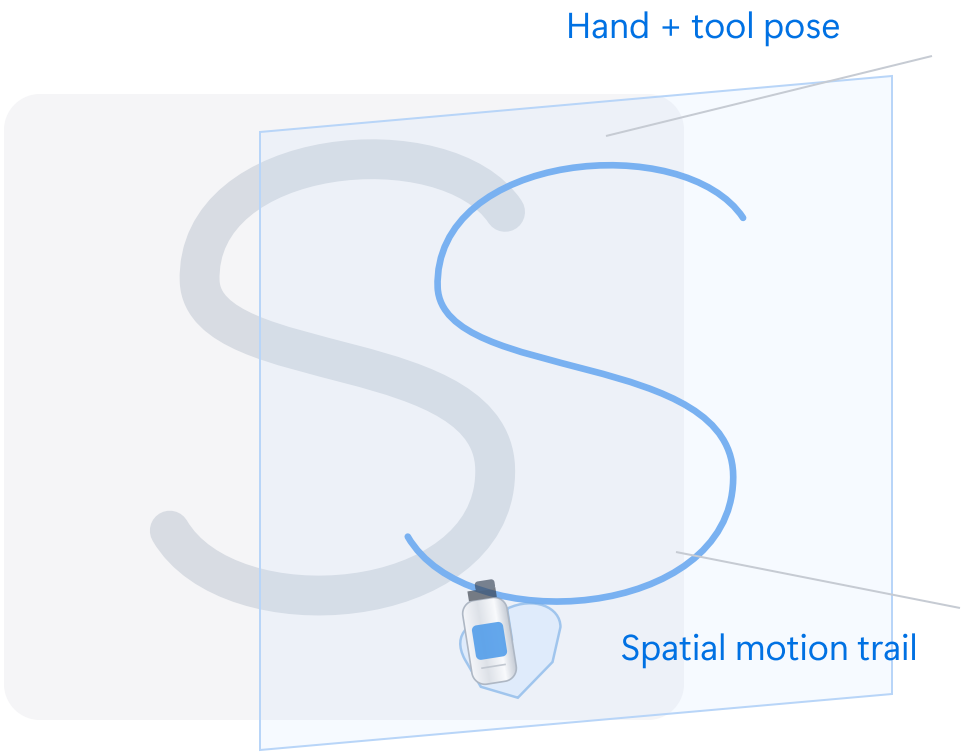
## Reproduce

Paint without the overlay.

## Remix

Make it your own.

# Guidance that steps back




## Observational learning

See stroke order
and timing.

## Scaffolding and fading

Full guide → faint trail
→ independent attempt.

## Feedback and reflection

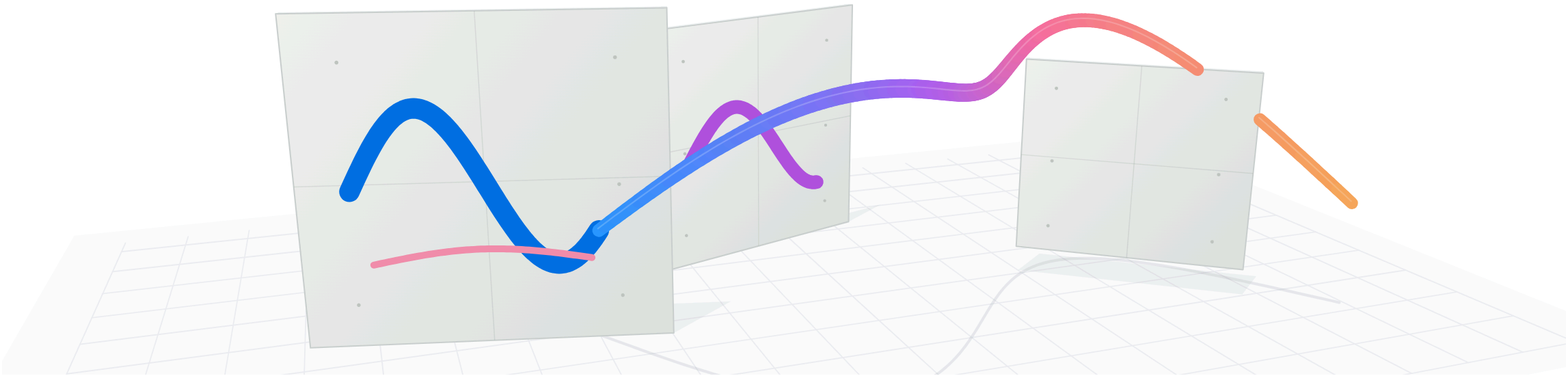
Distance cues during practice.
Path comparison afterward.

## Embodied interaction

Move, paint and follow cues
in the same space.

# Free Space to Practice

Shape your space. Create on your own terms.



Paint across surfaces and into open space

### Create surfaces

Add walls of different sizes.

### Arrange your space

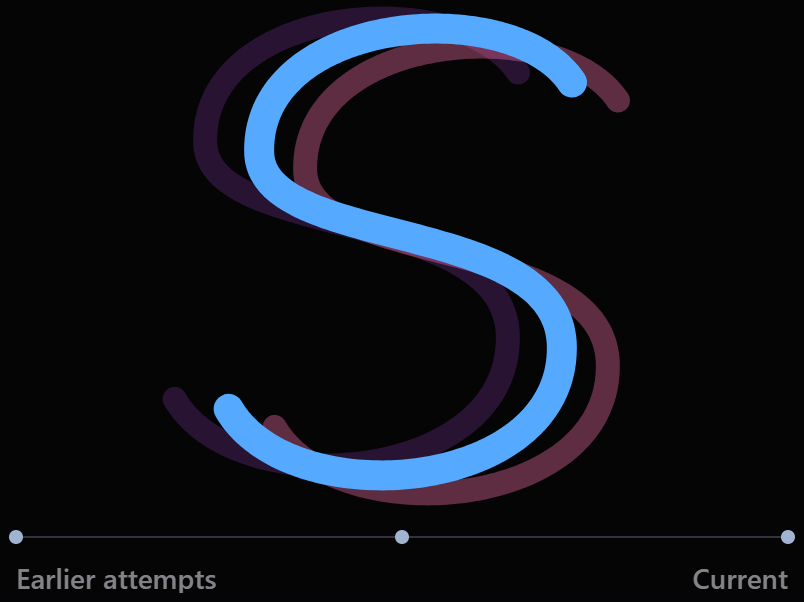
Move and rotate your layout.

### Go beyond walls

Extend your artwork into open space.

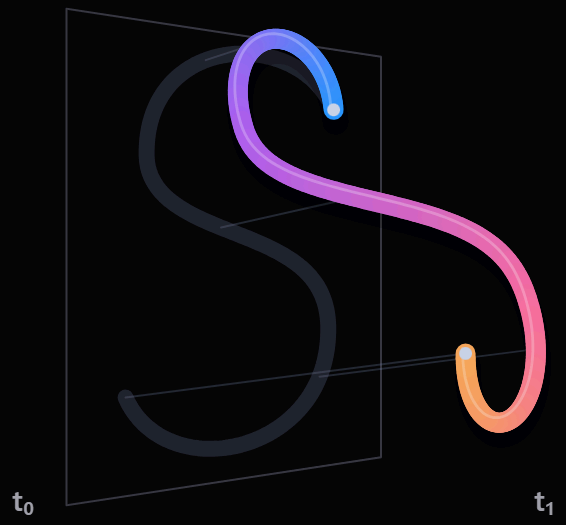
# Beyond Traditional 3D Art

Create with time, motion and recorded gestures.




## Temporal Layers

Past attempts.
One composition.




Time mapped to spatial depth

## Time-to-Space

Time becomes depth.
The stroke leaves the wall.

Beyond the MVP. Both reuse recorded gestures.

# One lesson.
# One semester.

**Smart Chen · Solo developer**
Experience: Unity XR, interaction, visual design and media.
Owns design, code, art direction, logging and evaluation.

Phase 1

## Foundations

Quest input + studio.
Create + arrange walls.

Phase 2

## Tools + lesson

Basic marker + roller.
Bubble S + lesson UI.

Phase 3

## Guided practice

Record + replay.
Ghost trail + fading cues.

Phase 4

## Test + polish

Unguided attempt + logs.
Usability + stable demo.

### Semester MVP

Editable walls. Three tools. One S lesson.
Guided practice and free creation.

### Stretch goals

Spatial painting. Temporal Layers.
Time-to-Space. Optional pilot.

# References

Research informs the design; outcomes remain untested.


**Han, Y., Ali, S. K. B. S., & Ji, L. (2022)**

Use of Observational Learning to Promote Motor Skill Learning in Physical Education: A Systematic Review.
*IJERPH, 19*(16), 10109. 18 included studies.

**Yu, X., Lee, B., & Sedlmair, M. (2024)**

Design Space of Visual Feedforward And Corrective Feedback in XR-Based Motion Guidance Systems.
*CHI 2024*.

**Infectious Ape. Kingspray Graffiti VR**

Official PC listing: spray, replay, projection and tutorials.
Features vary by platform.

**Krylon. How to Spray Paint**

Manufacturer guidance: technique and ventilation.

**Levac, D. E., Huber, M. E., & Sternad, D. (2019)**

Learning and transfer of complex motor skills in virtual reality: a perspective review.
*Journal of NeuroEngineering and Rehabilitation, 16*, 121.

**Wulf, G., Shea, C., & Lewthwaite, R. (2010)**

Motor skill learning and performance: a review of influential factors.
*Medical Education, 44*(1), 75–84.

**Open Brush**

Official project site. Room-scale 3D painting.

**Unity. XR input options**

Official Unity 6 documentation.
OpenXR, controller tracking and XR Interaction Toolkit.

# Discussion

Questions for the next iteration

## Why VR instead of a video?

Show tool motion at the wall.
Practice in the same space.

## Will this transfer to real paint?

Unknown: controllers lack material feedback.
Physical transfer needs a separate test.

## Why keep three tools?

Shared paint and recording systems.
Spray first; basic contact tools.

## What if the plan is too large?

Protect the spray lesson and logging.
Defer the pilot and temporal art.

## How will remix be assessed?

Reflection or artist feedback.
Path error measures copying.

## What is the first thing to validate?

Can beginners follow the cues
while seeing their own mark?

# Three tools. Different control.

Spray, draw and fill.




Montana BLACK 400 ml

## Spray can



Distance shapes the footprint




MTN Street Paint 15 mm

## Paint marker



Contact gives a crisp edge




Purdy Mini Roller System

## Paint roller



Broad passes build coverage

Controllers approximate motion; pressure, friction and weight are absent.